

$$R(A,B) = 80,000, S(A, C) = 20,000$$

- 1) 150 buffers means $M^2 = 22,500$. $B(R) > 22,500$, making two-pass sort-merge not an option. We can choose between hash join, since $B(R)+B(S) < M^2$, and block nested loop join. In this case, hash join would cost $3B(R) + 3B(S)$ which is 300,000 I/O accessors. Block nested loop join would obviously be much more costly than this, meaning we should use **two-pass Hash Join**.
- 2) 100 buffers means $M^2 = 10,000$. This means $B(R) > M^2$ and $B(S) > M^2$. We cannot use either hash join or sort-merge, meaning we must use **block nested loop** as the join method. The number of I/O accessors will be $B(R) + (B(R) / M - 2) * B(S) = 16,420,000$.
- 3) 320 buffers means $M^2 = 102,400$. Since $B(R) + B(S) < 102,400$, we can use optimized two-pass sort-merge join. The equation for that is $3B(R)+3B(S) = 300,000$ I/O accessors. The optimal method is **optimized two-pass sort-merge join**.